# Properties: Validity for $\nu \geq 0$, and Combining (7)–(8)

- In deriving (7), the sum was kept starting from $n = 0$ — valid only for $\nu > 0$. In (8), we only need $\nu \geq 0$.
- Taking $\nu = 0$ in (8): $J_0' = -J_1$.
- Taking $\nu = 0$ in (7): $J_0' = J_{-1}$. But $J_{-1} = -J_1$ (linear dependence, $m = 1$), so this gives the *same* relation as (8).
- Thus (7) is also valid for $\nu \geq 0$.
- $J_0(x)$ starts at $J_0(0) = 1$, decreases, crosses zero near $x \approx 2.405$, and continues oscillating with slowly *decreasing* amplitude as $x \to \infty$.
- $J_1(x)$ starts at $J_1(0) = 0$, rises to a maximum, then similarly oscillates with decreasing amplitude; its first positive zero is near $x \approx 3.832$.
- Expanding the derivatives in (7) and (8) via the product rule and simplifying gives

$$J_\nu'(x) + \frac{\nu}{x} J_\nu(x) = J_{\nu-1}(x), \qquad J_\nu'(x) - \frac{\nu}{x} J_\nu(x) = -J_{\nu+1}(x).$$



# Properties: The Two Master Recurrence Relations

- Adding and subtracting the two relations from the previous slide gives two more fundamental identities:

$$\boxed{J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x)} \tag{9}$$

$$\boxed{J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)} \tag{10}$$

- (9) expresses derivatives of $J_\nu$ purely in terms of neighbouring Bessel functions, without differentiating a series.
- (10) lets us climb to higher orders using only $J_0$ and $J_1$ and in practice, tables need only store these two. $J_{\nu+n}(x)$ can be found algorithmically from $J_\nu(x)$ and $J_{\nu+1}(x)$ for $n \in \mathbb{N}$.
- In particular, (10) is used to prove that $J_\nu(x)$ for $\nu = \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, \ldots$ are all **elementary** functions — expressible using rational functions and trigonometric functions.



# Relation Between $J_{1/2}(x)$ and $\sin x$

- From (3), taking $\nu = \frac{1}{2}$:

$$J_{1/2}(x) = \sqrt{x}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{2^{2n+1/2}n!\,\Gamma(n+3/2)} = \sqrt{\frac{2}{x}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2^{2n+1}n!\,\Gamma(n+3/2)}.$$

- Using $2^n n! = 2n(2n-2)\cdots 4\cdot 2$ and $2^{n+1}\Gamma(n+\frac{3}{2}) = (2n+1)(2n-1)\cdots 3\cdot 1\cdot\sqrt{\pi}$, we get

$$2^{2n+1}n!\,\Gamma(n+\tfrac{3}{2}) = (2n+1)!\sqrt{\pi}.$$

- Therefore,

$$\boxed{J_{1/2}(x) = \sqrt{\frac{2}{x}}\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)!\sqrt{\pi}} = \sqrt{\frac{2}{\pi x}}\sin x} \qquad (11)$$



# Relation Between $J_{-1/2}(x)$ and $\cos x$

- Multiplying (11) by $\sqrt{x}$ and differentiating with respect to $x$:

$$\left(\sqrt{x}\, J_{1/2}(x)\right)' = \sqrt{\frac{2}{\pi}} \cos x. \tag{12}$$

- Taking $\nu = \frac{1}{2}$ in (7):

$$\left(\sqrt{x}\, J_{1/2}(x)\right)' = \sqrt{x}\, J_{-1/2}(x). \tag{13}$$

- Combining (12) and (13):

$$\boxed{J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x} \tag{14}$$



# Spherical Bessel Functions

- Using (11) and (14) in the recurrence (10), $J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x) - J_{\nu-1}(x)$, we obtain:

$$J_{3/2}(x) = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left( \frac{\sin x}{x} - \cos x \right),$$

$$J_{-3/2}(x) = -\frac{1}{x} J_{-1/2}(x) - J_{1/2}(x) = -\sqrt{\frac{2}{\pi x}} \left( \frac{\cos x}{x} + \sin x \right),$$

$$J_{5/2}(x) = \frac{3}{x} J_{3/2}(x) - J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \left( \frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right),$$

$$J_{-5/2}(x) = -\frac{3}{x} J_{-3/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left( \frac{3 \cos x}{x^2} + \frac{3 \sin x}{x} - \cos x \right),$$

$$\ldots$$

- These are called **spherical Bessel functions**, as they arise in the solution of the wave equation in spherical coordinates.



# Problem : Computing $J_3(4)$ from $J_0, J_1$

- Using the recurrence $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$ twice, express $J_3(4)$ in terms of $J_0(4)$ and $J_1(4)$, and evaluate.
- With $n = 1$: $J_2(x) = \frac{2}{x} J_1(x) - J_0(x)$.
- With $n = 2$: $J_3(x) = \frac{4}{x} J_2(x) - J_1(x) = \left(\frac{8}{x^2} - 1\right) J_1(x) - \frac{4}{x} J_0(x)$.
- At $x = 4$: $J_0(4) = -0.397150$, $J_1(4) = -0.066043$, giving $J_2(4) = 0.364128$ and

$$J_3(4) = \left(\frac{8}{16} - 1\right)(-0.066043) - 1(-0.397150) = 0.430171.$$

- This matches the tabulated value of $J_3(4)$ (Check with Bessel's table).



# Problem : Evaluating $\int_0^2 xJ_0(x)\,dx$

- Using the identity $\dfrac{d}{dx}\left[xJ_1(x)\right] = xJ_0(x)$, evaluate $\displaystyle\int_0^2 xJ_0(x)\,dx$.
- Immediately,

$$\int_0^2 xJ_0(x)\,dx = \Big[xJ_1(x)\Big]_0^2 = 2J_1(2) - 0 \cdot J_1(0).$$

- Numerically, $J_1(2) = 0.576725$, so

$$\int_0^2 xJ_0(x)\,dx = 2(0.576725) = 1.153450.$$

- This matches direct **numerical integration** of $xJ_0(x)$ over $[0, 2]$.



# Problem

- **Q.** Evaluate the following in terms of $J_1$ and $J_2$:

$$\int J_3(x)\,dx.$$

- **Solution:** We can write $\int x^2(x^{-2}J_3(x))\,dx$ which is equivalent to

$$\int x^2(-x^{-2}J_2(x))'\,dx. \quad [\because (x^{-2}J_2(x))' = -x^{-2}J_3(x)]$$

- After integration by parts,

$$\begin{aligned}\int J_3(x)\,dx &= -x^2x^{-2}J_2(x) + \int 2x(x^{-2}J_2(x))\,dx \\ &= -J_2(x) + 2\int (x^{-1}J_2(x))\,dx \\ &= -J_2(x) - 2\int (x^{-1}J_1(x))'\,dx \\ &= -J_2(x) - 2x^{-1}J_1(x) + C.\end{aligned}$$



# Prove that $\int_0^1 (x - x^3) J_0(x)\, dx = 2J_2(1)$.

- **Solution:** LHS is

$$\begin{aligned}
\int_0^1 (x - x^3) J_0(x)\, dx &= \int_0^1 x J_0(x)\, dx - \int_0^1 x^3 J_0(x)\, dx \\
&= \int_0^1 x J_0(x)\, dx - \int_0^1 x^2 \big(x J_0(x)\big)\, dx. \\
&= \int_0^1 \big(x J_1(x)\big)'\, dx - \int_0^1 x^2 \big(x J_1(x)\big)'\, dx \\
&= x J_1(x)\big|_0^1 - x^3 J_1(x)\big|_0^1 + \int_0^1 2x\; x J_1(x)\, dx \\
&= J_1(1) - 0 - J_1(1) + 0 + 2\int_0^1 \big(x^2 J_2(x)\big)'\, dx \\
&= 2x^2 J_2(x)\big|_0^1 = 2J_2(1).
\end{aligned}$$



## Q. $\int x^4 J_1(x)\,dx = (4x^3 - 16x)J_1(x) - (x^4 - 8x^2)J_0(x) + C.$

- **Proof:** LHS $= \int x^4 J_1(x)\,dx = \int x^4 \left(-J_0'(x)\right) dx$, since $J_1(x) = -J_0'(x)$
- Thus, LHS $= -x^4 J_0(x) + \int 4x^3 J_0(x)\,dx$
-

$$= -x^4 J_0(x) + 4\int x^2 (xJ_1(x))'\,dx, \quad [\because xJ_0 = (xJ_1)']$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) - 8\int x^2 J_1(x)\,dx$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) + 8\int x^2 J_0'(x)\,dx$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) + 8x^2 J_0(x) - 16\int xJ_0(x)\,dx$$

$$= -(x^4 - 8x^2)J_0(x) + 4x^3 J_1(x) - 16\int (xJ_1)'\,dx$$

$$= (4x^3 - 16x)J_1(x) - (x^4 - 8x^2)J_0(x) + C = RHS$$



# The Generating Function

- For integer $n$, the entire family $\{J_n(x)\}_{n=-\infty}^{\infty}$ is packaged by a single closed-form expression:

$$\boxed{e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} J_n(x)\, t^n}$$

- That is, $J_n(x)$ is the coefficient of $t^n$ in the Laurent expansion (in $t$) of the left-hand side.
- This is the exact analogue of the generating function $(1-2xh+h^2)^{-1/2} = \sum P_n(x)h^n$ we used for Legendre polynomials — again a single function encoding an entire family.
- Nearly every identity for integer-order $J_n$ can be derived quickly by manipulating this generating function.



# Proof of Generating Function

Consider

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = e^{\frac{xt}{2}} e^{-\frac{x}{2t}}.$$

Using the exponential series,

$$e^{\frac{xt}{2}} = \sum_{r=0}^{\infty} \frac{1}{r!} \left(\frac{xt}{2}\right)^r$$

and

$$e^{-\frac{x}{2t}} = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!} \left(\frac{x}{2t}\right)^s.$$

Therefore,

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \left(\sum_{r=0}^{\infty} \frac{1}{r!} \left(\frac{xt}{2}\right)^r\right) \left(\sum_{s=0}^{\infty} \frac{(-1)^s}{s!} \left(\frac{x}{2t}\right)^s\right).$$



# Proof (ctn.)

Multiplying the two series gives

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{r,s\geq 0} \frac{(-1)^s}{r!s!} \left(\frac{x}{2}\right)^{r+s} t^{r-s}.$$

Put $n = r - s$ in the above. Then, for $n \geq 0$, we have $r = n + s$. Hence the coefficient of $t^n$ is

$$\sum_{s=0}^{\infty} \frac{(-1)^s}{(n+s)!s!} \left(\frac{x}{2}\right)^{n+2s}.$$

But, by the definition of $J_n(x)$,

$$J_n(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} \left(\frac{x}{2}\right)^{n+2s}.$$

Thus, for $n \geq 0$, the coefficient of $t^n$ is $J_n(x)$.



# Proof (ctn.)

For $n < 0$, write $n = -m$, where $m > 0$. Then $s = r + m$, and the coefficient of $t^{-m}$ is

$$\sum_{r=0}^{\infty} \frac{(-1)^{r+m}}{r!(r+m)!} \left(\frac{x}{2}\right)^{m+2r}.$$

Therefore,

$$[t^{-m}]e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = (-1)^m J_m(x).$$

Using the standard identity $J_{-m}(x) = (-1)^m J_m(x)$, we obtain

$$[t^{-m}]e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = J_{-m}(x).$$

Hence the coefficient of $t^n$ is $J_n(x)$ for every $n \in \mathbb{Z}$. Consequently, the generating function of the Bessel functions of the first kind $e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n$ is verified.



# Recurrence Formula for $J_n(x)$

- Earlier recurrence formulas can be obtained using the generating function for Bessel's function when $\nu = n$.
- $2nJ_n(x) = x[J_{n-1}(x) + J_{n+1}(x)]$ (Hint: differentiate the generating function w.r.t. $t$, equate the coefficient of $t^{n-1}$ – **Homework**.)
- $J_n'(x) = \frac{1}{2}[J_{n-1}(x) - J_{n+1}(x)]$ (Hint: differentiate the generating function w.r.t. $x$, equate the coefficient of $t^n$ – **Homework**.)
- From these two relations, we obtain

$$J_n'(x) = \frac{n}{x}J_n(x) - J_{n+1}(x),$$

$$J_n'(x) = -\frac{n}{x}J_n(x) + J_{n-1}(x).$$

